

Lab Assignments:Case study one

- 1) Consider the dataset [stocksreturn](#).
 - a) Conduct factor analysis by using two of these six rotation methods [varimax](#), [quartimax](#), [equamax](#), [oblimin](#), [oblimax](#), [target_oblique](#), and compare results (via plot)
 - b) Try even Principal Component Analysis method, can you achieve the same result as Factor Analysis ?
 - c) Collect your own data by looking at the last 20 days at the Stockholm's Nasdaq index for these companies: ABB, SAAB, Volvo, Atlas Copco; NCC, JM, Skanska; SEB, Nordea bank, Handelsbanken. Then carry out a factor analysis for your data.
- 2) Modify the code in [genLDdata.m](#) so that the sphere and line segments are well separated thereafter study both global and local dimensions. Can you obtain a better classification of dataset in various dimensions compared with the results in Example 2.10 ?
- 3) Inspect the code in [intrinsic_dim.m](#) and test if changes in some parameters would yield better results
 - a) in case '[NearNbDim](#)' the parameters are [k1](#), [k2](#) with default value [k1 = 6](#), [k2 = 12](#).
 - b) in case '[PackingNumbers](#)' the parameters are [r\(1\)](#), [r\(2\)](#) and [max_iter](#) with default values, [0.1](#), [0.5](#) and [20](#).Estimate the dimension of helix with noise level $\sigma = 0.1$

On Wednesday Feb 4 , 2026, the time 14:15 – 15:00 is devoted to demonstration of solutions in classroom.

Three students will be randomly chosen to show their results.

Every student should also upload her/his report in Canvas at the latest 11:59 PM Feb 4, 2026.

Attendance is mandatory